Module/Pathway/Taxon Review: Galactose Catabolism (Leloir Pathway) in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Target bucket:** KEGG `ppu00052` "Galactose metabolism" **Module area:** other_kegg_pathway **Curation verdict: GAP / module_needs_revision** — the catabolic Leloir pathway is **not satisfiable** in KT2440.

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1. Executive Summary

The four-reaction Leloir pathway (GALM → GALK → GALT → GALE) that converts extracellular D-galactose into glucose-1-phosphate is **not encoded** in the native genome of *Pseudomonas putida* KT2440. The two committed catabolic steps — **galactokinase (GALK, EC 2.7.1.6)** and **galactose-1-phosphate uridylyltransferase (GALT, EC 2.7.7.12)** — have **no corresponding gene**. This is confirmed at two independent levels of KEGG annotation: EC-based queries (`link/ppu/ec:2.7.1.6` and `link/ppu/ec:2.7.7.12`) return zero genes, and KEGG-ortholog (KO) queries (galactokinase K00849; Gal-1-P uridylyltransferase K00965) are likewise absent. Because a linear catabolic pathway is broken the moment a committed step is missing, the module is unsatisfiable and should be curated as a **gap**, not as "covered."

The eight candidate genes assigned to `ppu00052` in the local metadata are almost entirely **central hexose-phosphate and nucleotide-sugar interconversion enzymes** that KEGG's map-based bucketing sweeps into the "Galactose metabolism" map. Only two of them belong to genuine Leloir enzyme families: **GALM** (aldose 1-epimerase/mutarotase, PP_1165) and **GALE** (UDP-glucose 4-epimerase, PP_3129 plus paralog PP_0501). Even GALE, when present, operates in KT2440's **UDP-sugar / LPS biosynthesis**, not in a catabolic UDP-glucose recycling loop, because the GALT reaction that would feed it UDP-galactose does not occur. The remaining candidates — `glk` (glucokinase, EC 2.7.1.2), `galU` (UTP-glucose-1-P uridylyltransferase, EC 2.7.7.9), `pgm`, `cpsG`, and `algC` (phosphoglucomutases/phosphomannomutases) — are generic and must **not** be counted toward the committed Leloir steps.

The two candidates most likely to trigger a false "covered" call are `glk` and `galU`, because of **EC/GO term breadth** ("sugar kinase", "uridylyltransferase"). Neither performs the Leloir chemistry: glucokinase phosphorylates glucose at C6 (not galactose at C1), and galU is a biosynthetic UDP-glucose pyrophosphorylase (glucose-1-P + UTP → UDP-glucose), not the Gal-1-P/UDP-glucose exchange catalyzed by GALT. The negative conclusion is corroborated by two independent metabolic-engineering studies that had to **install** galactose catabolism into KT2440 — one integrating the complete Leloir operon *galETKM* [PMID: 40691973], the other integrating a **De Ley–Doudoroff** oxidative pathway [PMID: 31890023] — direct experimental proof that the wild-type strain cannot catabolize galactose by either route.

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2. Target-Organism Pathway Definition

What the module is (working scope)

The **Leloir pathway of galactose catabolism** is a four-reaction, cytoplasmic route that channels free D-galactose into central carbon metabolism as glucose-1-phosphate:

1. **Mutarotation** — β-D-galactose ⇌ α-D-galactose (GALM, aldose 1-epimerase, EC 5.1.3.3)
2. **Phosphorylation (committed)** — α-D-galactose + ATP → α-D-galactose-1-phosphate + ADP (GALK, galactokinase, EC 2.7.1.6)
3. **Uridylyl transfer (central)** — Gal-1-P + UDP-glucose → glucose-1-phosphate + UDP-galactose (GALT, EC 2.7.7.12)
4. **UDP-sugar recycling** — UDP-galactose ⇌ UDP-glucose (GALE, UDP-glucose 4-epimerase, EC 5.1.3.2)

The pathway is **catalytic in its UDP-sugar co-substrate**: GALE regenerates the UDP-glucose that GALT consumes, so the net transformation is galactose → glucose-1-phosphate. This is the defining feature that distinguishes true catabolic Leloir flux from the isolated presence of GALE, which many organisms carry purely for **nucleotide-sugar (UDP-galactose) biosynthesis** feeding glycan/LPS assembly.

Boundaries — what to keep separate

- **Nucleotide-sugar biosynthesis / LPS and EPS precursor supply.** The presence of GALE (and galU, pgm) alone indicates the cell can make and interconvert UDP-glucose/UDP-galactose for polysaccharide synthesis — this is **not** galactose catabolism. In

Acidithiobacillus ferrooxidans, galE/galU/pgm/galT-like genes were shown to supply EPS precursors [PMID: 15932984]. The same enzymes in KT2440 serve biosynthesis.

- **De Ley–Doudoroff oxidative galactose pathway.** An alternative, Leloir-independent catabolic route (galactose → galactonate → 2-keto-3-deoxygalactonate → pyruvate + glyceraldehyde-3-P via dgoD/dgoK/dgoA). This must be checked separately as a lineage-specific alternative — and in KT2440 it, too, is absent (see Finding F005).
- **Broad KEGG overview maps** (e.g., ppu01100 "metabolic pathways", ppu00520 "amino sugar and nucleotide sugar metabolism") should not be conflated with the specific Leloir catabolic module.

Alternate names / database definitions

- KEGG map: `ppu00052` "Galactose metabolism" (a broad map, **not** a Leloir-specific module).
- Gene symbols across bacteria: `galM` (GALM), `galK` (GALK), `galT` (GALT), `galE` (GALE), often organized as a `gal` operon (e.g., `galKETRM` in *Lactobacillus casei* [PMID: 9603808]; `galKT` in *Streptococcus pneumoniae* [PMID: 26544195]).
- The "Leloir pathway" name is used interchangeably with "galactose catabolism via galactose-1-phosphate."

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3. Expected Step Model and Satisfiability in KT2440

Step	Enzyme (family)	EC	KO	Expected gene	KT2440 status	Candidate
1. Mutarotation	GALM (aldose 1-epimerase)	5.1.3.3	K01785	galM	**candidate_uncertain**	PP_1165 (generic; KEGG NAME "conserved protein of unknown function")
2. Phosphorylation (committed)	GALK (galactokinase)	2.7.1.6	K00849	galK	**GAP (absent)**	none (glk PP_1011 = EC 2.7.1.2 glucokinase)
3. Uridylyl transfer (central)	GALT	2.7.7.12	K00965	galT	**GAP (absent)**	none (galU PP_3821 = EC 2.7.7.9)
4. UDP-sugar recycling	GALE (UDP-glc 4-epimerase)	5.1.3.2	K01784	galE	**present but repurposed** (biosynthesis)	PP_3129, paralog PP_0501

Conclusion: With **two committed catabolic steps absent (GALK, GALT)**, the linear catabolic module is broken and **unsatisfiable**. GALM is present only as a generic aldose 1-epimerase of uncertain galactose specificity; GALE is present but functions in nucleotide-sugar/LPS biosynthesis rather than closing a catabolic recycling loop (there is no UDP-galactose being produced by GALT to recycle).

4. Candidate Genes and Evidence

The local metadata lists **8 candidate genes**. Below is the KEGG-ortholog-resolved assessment of each (Finding F002). Only PP_1165 and PP_3129/PP_0501 map to genuine Leloir enzyme families; the rest are central hexose-phosphate / nucleotide-sugar metabolism.

Gene	Locus	UniProt	KEGG KO / EC	True function	Leloir role?
PP_0501	PP_0501	Q88QJ1	K01784 / EC 5.1.3.2	UDP-glucose 4-epimerase (GALE paralog)	Step 4 (biosynthetic)
galE	PP_3129	Q88I72	K01784 / EC 5.1.3.2	UDP-glucose 4-epimerase (GALE)	Step 4 (biosynthetic)
PP_1165	PP_1165	Q88NP2	K01785 / EC 5.1.3.3	Aldose 1-epimerase (mutarotase)	Step 1 (uncertain)
glk	PP_1011	Q88P42	K00845 / EC 2.7.1.2	**Glucokinase** (not galactokinase)	**none — over-propagation risk**
galU	PP_3821	Q88GA4	K00963 / EC 2.7.7.9	**UDP-glucose pyrophosphorylase** (not GALT)	**none — over-propagation risk**
pgm	PP_3578	Q88GY7	K01835 / EC 5.4.2.2	Phosphoglucomutase	none (central)
cpsG	PP_1777	Q88LZ9	K15778 / EC 5.4.2.8+5.4.2.2	Phosphomannomutase/ PGM	none (central)
algC	PP_5288	Q88C93	K15778 / EC 5.4.2.8+5.4.2.2	Phosphomannomutase/ PGM (alginate/LPS)	none (central)

High-confidence assessments

- **GALE — PP_3129 (galE) + PP_0501 (paralog).** Both are bona fide UDP-glucose 4-epimerases (K01784, EC 5.1.3.2). Their presence is real and high-confidence at the *enzyme* level, but the **biological role in KT2440 is nucleotide-sugar/LPS biosynthesis**, not catabolic UDP-glucose recycling. In a true Leloir pathway, GALE recycles the UDP-galactose that GALT produces; with GALT absent, there is no catabolic UDP-galactose to recycle, so GALE cannot be counted as "covering" a catabolic step. Curation-relevant caveat: **paralog ambiguity** — two GALE-family genes exist and should not be double-counted, and neither implies catabolism. (GALE-family genes function in LPS/precursor biosynthesis across bacteria, e.g., orfH8 complementing a *Salmonella* galE mutant [PMID: 10858186]; galE in the *A. ferrooxidans* EPS-precursor cluster [PMID: 15932984].)

- **GALM — PP_1165.** Assigned K01785 (aldose 1-epimerase, EC 5.1.3.3), but the KEGG NAME is "conserved protein of unknown function." It is a **generic aldose 1-epimerase**, not demonstrated to be galactose-specific. Aldose 1-epimerases act on multiple aldohexoses; mutarotase activity alone does not establish a committed galactose-catabolic role. Verdict: **candidate_uncertain**.

Likely over-propagations (do NOT count as Leloir steps) — Finding F003

- **glk — PP_1011 (glucokinase, K00845, EC 2.7.1.2).** Mechanistically distinct from galactokinase: glucokinase phosphorylates **glucose at C6**, whereas GALK phosphorylates **galactose at C1** (EC 2.7.1.6). Broad "sugar kinase" GO/EC terms can cause this to be miscounted as satisfying the committed phosphorylation step. It does **not**.
- **galU — PP_3821 (UTP-glucose-1-P uridylyltransferase, K00963, EC 2.7.7.9).** A **biosynthetic UDP-glucose pyrophosphorylase** (glucose-1-P + UTP → UDP-glucose + PPi). This is the biosynthetic uridylyltransferase — completely different from GALT (EC 2.7.7.12), which exchanges UMP between Gal-1-P and UDP-glucose. Because both carry "uridylyltransferase" in their names, galU is the single most likely gene to be **falsely** mapped to the central Leloir step. It does **not** perform GALT chemistry.
- **pgm (PP_3578), cpsG (PP_1777), algC (PP_5288).** Phosphoglucomutase / phosphomannomutase enzymes of central carbon and nucleotide-sugar metabolism. They convert glucose-1-P ⇌ glucose-6-P (downstream of, or parallel to, the Leloir output) and support LPS/alginate biosynthesis. Not Leloir catabolic steps.

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5. Gaps, Ambiguities, and Likely Over-Annotations

Confirmed gaps (committed steps absent)

Two orthogonal KEGG queries confirm the absence of the committed catabolic steps (Findings F001, F005):

- **EC level:** `link/ppu/ec:2.7.1.6` (galactokinase) → **0 genes**; `link/ppu/ec:2.7.7.12` (Gal-1-P uridylyltransferase) → **0 genes**.
- **KO level:** galactokinase (K00849) → **absent**; Gal-1-P uridylyltransferase (K00965) → **absent**.

The De Ley–Doudoroff alternative is also absent

A lineage-specific oxidative alternative to Leloir was checked and ruled out at the KO level (Finding F005):

- galactonate dehydratase **dgoD (K01684) — absent**
- 2-oxo-3-deoxygalactonate kinase **dgoK (K00875) — absent**
- KDPG-galactonate aldolase **dgoA (K01631) — absent**
- galactose dehydrogenase **(K18649) — absent**

So KT2440 lacks **both** known galactose-catabolic routes natively.

Over-annotation risks summary

Risk	Gene	Why it's misleading
Kinase over-propagation	glk (PP_1011)	"sugar kinase"/EC 2.7.1.x breadth; is glucokinase not galactokinase
Uridylyltransferase over-propagation	galU (PP_3821)	shares "uridylyltransferase" with GALT; is biosynthetic UDP-Glc pyrophosphorylase
Map-bucket inflation	pgm, cpsG, algC	central mutases swept into ppu00052 broad map
Paralog double-count	PP_3129 + PP_0501	two GALE genes; biosynthetic, not catabolic
GALM specificity	PP_1165	generic aldose 1-epimerase, "unknown function"

6. Module and GO-Curation Recommendations

Per-step module status (Finding F004)

Module step	Recommended status	Rationale
1. GALM (mutarotation)	**candidate_uncertain**	PP_1165 generic aldose 1-epimerase, not proven galactose-specific
2. GALK (phosphorylation)	**gap**	no gene; EC 2.7.1.6 and K00849 empty
3. GALT (uridylyl transfer)	**gap**	no gene; EC 2.7.7.12 and K00965 empty
4. GALE (recycling)	**present-but-repurposed** (biosynthesis; not catabolic)	PP_3129 + PP_0501 function in nucleotide-sugar/LPS biosynthesis
Whole module	**module_needs_revision / gap**	two committed catabolic steps absent → unsatisfiable

Module boundary judgment

The generic Leloir module boundaries are **biochemically correct** but the KEGG `ppu00052` bucket is **too broad** for KT2440: it aggregates central hexose-phosphate and nucleotide-sugar enzymes (glk, galU, pgm, cpsG, algC) that create a false impression of pathway completeness. Recommendation:

- **Do not mark the catabolic Leloir module "covered."** Mark it a **gap** with `module_needs_revision`.
- Explicitly annotate glk and galU as **excluded** from the GALK/GALT steps (over-propagation guards).
- Consider splitting the KEGG map bucket so that biosynthetic UDP-sugar interconversion (GALE, galU, pgm) is tracked separately from catabolic galactose utilization.

GO-curation notes

- Avoid transferring GO:galactokinase activity or GO:UDP-glucose:hexose-1-P uridylyltransferase activity to any KT2440 gene — no gene qualifies.

- PP_1165: retain generic **aldose 1-epimerase activity** (GO:0004034) but **do not** upgrade to galactose-specific catabolic annotation without direct assay.
- PP_3129/PP_0501: annotate as **UDP-glucose 4-epimerase (GO:0003978)** in the **nucleotide-sugar biosynthesis** context, not galactose catabolism.

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7. Genes to Promote to Full `fetch-gene` Review

1. **PP_1165 (putative GALM).** Highest priority. It is the only candidate for step 1 and is annotated "conserved protein of unknown function." A full review should establish substrate specificity (galactose vs. glucose/other aldoses) and genomic context (is it near any cryptic catabolic genes?).
2. **PP_3821 (galU) and PP_1011 (glk).** Promote to confirm the **over-propagation exclusion** — verify EC 2.7.7.9 and EC 2.7.1.2 assignments respectively and formally record that they do **not** satisfy the GALT/GALK steps. This protects against automated recount.
3. **PP_3129 (galE)** + **PP_0501.** Promote to resolve **paralog roles** and confirm the biosynthetic (LPS/nucleotide-sugar) rather than catabolic context; ensure they are not double-counted.

Lower priority (central metabolism, unlikely to change verdict): pgm (PP_3578), cpsG (PP_1777), algC (PP_5288).

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8. Mechanistic Model / Interpretation

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            CANONICAL LELOIR (what the module expects)
b-D-Gal --GALM--> a-D-Gal --GALK--> Gal-1-P --GALT--> Glc-1-P
 (5.1.3.3)          (2.7.1.6)          |    (2.7.7.12)
                                       |  <---- UDP-Glc
                                       v
                                  UDP-Gal --GALE--> UDP-Glc (recycled)
                                            (5.1.3.2)

            WHAT KT2440 ACTUALLY HAS
b-D-Gal --[PP_1165?]--> a-D-Gal --?--X   GALK ABSENT
                                         GALT ABSENT
                                         -----------------------
UDP-Glc <==GALE (PP_3129/PP_0501)==> UDP-Gal   [biosynthesis only]
     ^                                    for LPS / nucleotide-sugars
     |
 galU / pgm / algC / cpsG   (central hexose-P & UDP-sugar supply)
```

The KT2440 genome retains the **biosynthetic/recycling arm** (GALE) and a possible **mutarotase** (PP_1165), but is missing the **entrance machinery** — the committed galactokinase and the central uridylyltransferase — that would let free galactose flow into central metabolism. Consequently the "recycling" arm has nothing to recycle from a catabolic standpoint: GALE serves anabolism (making UDP-galactose for glycoconjugates), which is the opposite direction of Leloir catabolism.

This interpretation is not merely computational. Two metabolic-engineering studies independently demonstrate that wild-type KT2440 cannot use galactose and had to be **retrofitted**:

- **Leloir retrofit:** engineers integrated the complete Leloir operon *galETKM* (including galK and galT) plus a lactose permease into the KT2440 chromosome to enable lactose/galactose utilization [PMID: 40691973] — direct evidence the native strain lacks galK and galT.
- **De Ley–Doudoroff retrofit:** a separate study installed an entirely different (oxidative) galactose-catabolic pathway into *P. putida* [PMID: 31890023] — indicating that neither the Leloir nor a native oxidative route works in the wild type.

Both results are **direct, target-organism evidence** and converge on the same conclusion as the KEGG EC/KO analysis.

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9. Evidence Base

PMID	Organism	Relevance	Supports/Challenges
PMID: 40691973	*P. putida* KT2440	Engineers integrated Leloir *galETKM* + *lacY* into KT2440 chromosome to enable galactose/ lactose use	**Supports** (direct): native strain lacks galK/galT
PMID: 31890023	*P. putida*	Integrated a De Ley–Doudoroff galactose pathway into the chromosome	**Supports** (direct): no functional native galactose catabolism
PMID: 15932984	*A. ferrooxidans*	gal cluster (galE/galK/pgm/galM + galU, galT-like) supplies EPS/UDP-sugar precursors	**Context**: GALE/galU serve biosynthesis, not always catabolism
PMID: 9603808	*L. casei*	Canonical gal operon galKETRM with galK/galT/galE enzyme activities	**Reference model** of a complete Leloir operon (contrast to KT2440)
PMID: 12839781	*L. raffinolactis*	aga-galKT operon; galK+galT required for galactoside catabolism	**Reference model**: committed galK/galT define catabolic capability
PMID: 26544195	*S. pneumoniae*	galK required for growth on galactose	**Supports**: GALK is the committed, indispensable catabolic step
PMID: 10858186	*L. borgpetersenii*	GalE-like gene complements *Salmonella* galE in LPS biosynthesis	**Context**: GALE-family = biosynthesis (LPS), not catabolism
PMID: 10993714	*S. cerevisiae*	GALT catalyzes second step of Leloir, following GALK, preceding GALE	**Reference** for canonical step order and GALT centrality

Verified direct-organism quotes: - *"the expression of β-galactosidase gene lacZ on a plasmid was accompanied with integration of galactose Leloir pathway genes galETKM and lactose permease gene lacY into the chromosome of KT2440"* [PMID: 40691973]. - *"we integrated a De Ley-Doudoroff catabolic pathway for galactose catabolism into the chromosome of"* [PMID: 31890023].

10. Limitations and Knowledge Gaps

- **Absence of evidence vs. evidence of absence.** The GALK/GALT gaps rest on KEGG EC/KO annotation completeness. While corroborated by two orthogonal query types (EC and KO) and by direct engineering studies, a divergent, unannotated galactokinase or uridylyltransferase cannot be fully excluded by KEGG alone. A HMM/profile search against the KT2440 proteome would strengthen the negative.
- **GALM specificity unresolved.** PP_1165's true substrate range (galactose vs. glucose vs. other aldoses) is not experimentally established; "aldose 1-epimerase" is a broad activity.
- **GALE directionality/role inferred.** The assignment of PP_3129/PP_0501 to biosynthesis is inferred from pathway logic and homology (no galactose catabolic sink exists), not from a KT2440-specific flux measurement.
- **No transporter analysis.** Even a complete Leloir pathway requires a galactose uptake system; this review did not assess galactose transport, another potential bottleneck.
- **Species transfer.** The reference operon studies (*L. casei*, *L. raffinolactis*, *S. pneumoniae*, *A. ferrooxidans*, *S. cerevisiae*) are used only to define the canonical pathway and enzyme roles; they are **not** transferred to KT2440. The KT2440-specific conclusions rest on KEGG ppu queries and the two *P. putida* engineering papers.

11. Proposed Follow-up Experiments / Actions

1. **Profile-HMM confirmation of the gaps.** Run galactokinase (GHMP kinase, GALK) and GALT (histidine-triad Gal-1-P uridylyltransferase) HMMs against UP000000556 to formally exclude a divergent, unannotated ortholog. Expected: no hit — would harden the "gap" call.
2. **Promote PP_1165 to full `fetch-gene` review.** Determine galactose specificity (literature, structural homology, operon context) to decide GALM = candidate_uncertain vs. covered vs. not_expected.
3. **Lock over-propagation guards.** In the module document, explicitly record that PP_1011 (glk, EC 2.7.1.2) and PP_3821 (galU, EC 2.7.7.9) do **not** satisfy GALK/GALT, to prevent automated recount.
4. **Annotate GALE context.** Reassign PP_3129/PP_0501 GO/pathway context to nucleotide-sugar/LPS biosynthesis; flag the paralog pair to avoid double counting.

5. **Curation action:** mark KEGG `ppu00052` catabolic Leloir module as **gap** / **module_needs_revision**; request a module-boundary split separating biosynthetic UDP-sugar interconversion from catabolic galactose utilization.
6. **Optional wet-lab confirmation:** growth assay of wild-type KT2440 on D-galactose as sole carbon source (expected: no growth), matching the phenotype implied by the two engineering studies.

Bottom line for curators

> **Catabolic Leloir module in *P. putida* KT2440 = GAP (unsatisfiable).** GALK (EC 2.7.1.6) and GALT (EC 2.7.7.12) are absent at both EC and KO levels; the De Ley–Doudoroff alternative is also absent. GALM (PP_1165) = candidate_uncertain; GALE (PP_3129/PP_0501) present but repurposed for biosynthesis. glk and galU are over-propagation traps and must be excluded from the committed steps.
